Computational Economics

Lecture: Value Functions and the Bellman Equation

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Lecture Overview

- **The Value Function**
  - Policy rollouts, reward streams, and discounted returns
  - Expected return from a starting state
- **The Bellman Equation**
  - Recursive returns and policy evaluation
  - The linear system for a fixed policy
  - Optimal value and the choice of an action
- **Two Economic Models**
  - Job search: the McCall model
  - Consumption and saving with uncertain income

Recap: A Markov Decision Process

A time-homogeneous discounted **MDP** is specified by $\big(\mathcal{S}, \{\mathcal{A}(s)\}_{s\in\mathcal{S}}, P, r, \beta\big)$.

- **State space $\mathcal{S}$:** free cells $\mathcal{S} = \{\mathrm{S}, \mathrm{G}, 1, \ldots, 45\}$, including G. Walls are excluded.
- **Actions $\mathcal{A}(s)$:** up, down, left, right.
- **Transition kernel $P(s' \mid s, a)$:** probability 1 for the cell reached by action a, and 0 for all other cells. For example, $P(21 \mid \mathrm{S}, \text{down}) = 1$. A wall or boundary keeps the current cell; G is absorbing.
- **Mean reward $r(s, a)$:** -1 per move from outside G (including entry into G); zero at G.
- **Discount factor $\beta \in (0, 1)$:** weights future rewards.

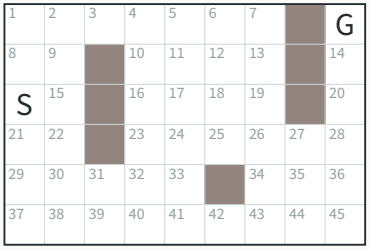


Figure 1: Maze navigation.

Recap: A Policy Chooses an Action at Each State

A **stationary deterministic policy** is a rule σ with $\sigma(s) \in \mathcal{A}(s)$: at cell s, take action $a = \sigma(s)$.

$\sigma_A = (\text{S} \to \text{down},\ \text{cell } 21 \to \text{down},\ \text{cell } 29 \to \text{right}, \ldots)$

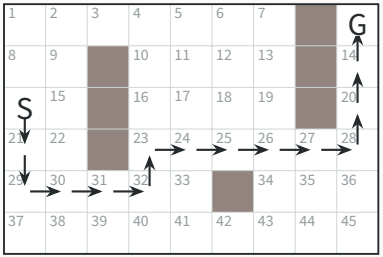


Figure 2: Policy σ_A.

$\sigma_B = (\text{S} \to \text{up},\ \text{cell } 8 \to \text{up},\ \text{cell } 1 \to \text{right}, \ldots)$

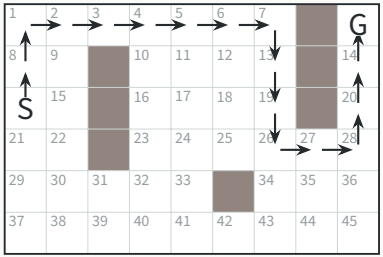


Figure 3: Policy σ_B.

Rollouts: Simulating a Policy

In reinforcement learning, a **rollout** follows a policy from a starting state, generating a **trajectory** of states, actions, and rewards.

To simulate a rollout from $s_t = s$, repeat for $\tau = 1, 2, \ldots$:

$$a_{t+\tau-1} = \sigma(s_{t+\tau-1}), \qquad s_{t+\tau} \sim P(\cdot \mid s_{t+\tau-1}, a_{t+\tau-1}).$$

The environment also generates reward $R_{t+\tau}$, whose conditional mean is $r(s_{t+\tau-1}, a_{t+\tau-1})$.

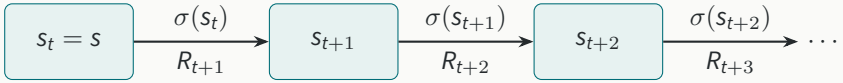


Figure 4: A rollout under policy σ.

The associated **reward stream** is $R_{t+1}, R_{t+2}, \ldots$. Rollouts can differ when transitions or rewards are random.

The Return of a Rollout

The **discounted accumulated reward**, or **return**, assigns a number to the reward stream of a rollout starting at date t:

$$G_t = R_{t+1} + \beta R_{t+2} + \cdots + \beta^{\tau-1} R_{t+\tau} + \cdots = \sum_{\tau=1}^{\infty} \beta^{\tau-1} R_{t+\tau}.$$

- The first reward has weight 1; the reward τ steps ahead has weight $\beta^{\tau-1}$.
- One rollout gives one realized return. Random transitions or rewards can produce different returns under the same policy.
- To evaluate a policy from a given state, we consider the expected return over its possible rollouts.

Discounting and a Finite Return

Let $\bar{r} < \infty$ be a **uniform bound on absolute rewards**: $|R_{t+j}| \leq \bar{r}$ for every $j \geq 1$.

For any rollout truncated after τ periods,

$$\left|G_t^{(\tau)}\right| = \left|\sum_{j=1}^{\tau} \beta^{j-1} R_{t+j}\right| \leq \sum_{j=1}^{\tau} \beta^{j-1} |R_{t+j}|$$
$$\leq \bar{r} \sum_{j=1}^{\tau} \beta^{j-1} = \frac{1-\beta^{\tau}}{1-\beta}\, \bar{r}.$$

As $\tau \to \infty$, $\beta^{\tau} \to 0$, so the infinite-horizon return satisfies

$$|G_t| \leq \lim_{\tau \to \infty} \frac{1-\beta^{\tau}}{1-\beta}\, \bar{r} = \frac{\bar{r}}{1-\beta}.$$

Standing Assumptions

Unless explicitly stated otherwise, we assume:

- **Time-homogeneous dynamics.** For a given state and action, the next-state probabilities and mean reward are the same at every date:

$$P_t(s' \mid s, a) = P(s' \mid s, a), \qquad r_t(s, a) = r(s, a).$$

- **A stationary policy.** The action rule depends on the current state, not the date:

$$\sigma_t(s) = \sigma(s).$$

  Actions can still change when the state changes.
- We therefore omit date subscripts on P, r, and σ. Over an infinite horizon, the expected return from a state also does not depend on the starting date; we write $V^{\sigma}(s)$.

The Value Function of a Policy

The **state-value function under policy** σ is the expected return over rollouts starting at s:

$$V^{\sigma}(s) = \mathbb{E}_{\sigma}[G_t \mid s_t = s] = \mathbb{E}_{\sigma}\left[\sum_{\tau=1}^{\infty} \beta^{\tau-1} R_{t+\tau} \,\middle|\, s_t = s\right].$$

- The subscript σ in $\mathbb{E}_{\sigma}$ means that the expectation is over rollouts following this particular policy σ.
- G_t is the return of one rollout. $V^{\sigma}(s)$ averages over all possible rollouts from s under σ.
- Since $r(s, a)$ is the conditional mean reward, boundedness also gives

$$V^{\sigma}(s) = \mathbb{E}_{\sigma}\left[\sum_{\tau=1}^{\infty} \beta^{\tau-1} r(s_{t+\tau-1}, \sigma(s_{t+\tau-1})) \,\middle|\, s_t = s\right].$$

The Return Has a Recursive Form

Along a rollout, the return from date t is:

$$G_t = R_{t+1} + \beta R_{t+2} + \beta^2 R_{t+3} + \cdots .$$

Separate the first reward from all later rewards:

$$\begin{aligned} G_t &= R_{t+1} + \beta\big(R_{t+2} + \beta R_{t+3} + \cdots\big) \\ &= R_{t+1} + \beta G_{t+1}. \end{aligned}$$

- G_{t+1} is the return from the remainder of the same rollout.
- From the next state s', the expected remaining return is

$$\mathbb{E}_\sigma[G_{t+1} \mid s_{t+1} = s'] = V^\sigma(s').$$

This lets us replace an infinite reward sequence with a one-step equation.

The Bellman Equation for a Fixed Policy

Average the rollout identity $G_t = R_{t+1} + \beta G_{t+1}$ conditional on $s_t = s$:

$$\begin{aligned} V^\sigma(s) &= \mathbb{E}_\sigma[R_{t+1} + \beta G_{t+1} \mid s_t = s] \\ &= r(s, \sigma(s)) + \beta \mathbb{E}[V^\sigma(s_{t+1}) \mid s_t = s,\ a_t = \sigma(s)]. \end{aligned}$$

The second equality uses the law of iterated expectations and the Markov structure.

For a finite state space, the **policy Bellman equation** is

$$V^\sigma(s) = r(s, \sigma(s)) + \beta \sum_{s' \in \mathcal{S}} P(s' \mid s, \sigma(s)) V^\sigma(s').$$

Current expected reward plus discounted expected continuation value. For a continuous state space, replace the sum by an integral with respect to $P(ds' \mid s, \sigma(s))$.

Policy Evaluation as a Linear System

For a finite state space, fix a policy σ and define

$$P_{ij}^{\sigma} = P(j \mid i, \sigma(i)), \qquad r_i^{\sigma} = r(i, \sigma(i)).$$

- Row i of P^{σ} gives next-state probabilities when we follow σ from state i.
- Entry i of r^{σ} is the current expected reward at that state.

Treat V^{σ} and r^{σ} as column vectors. The policy Bellman equations become

$$V^{\sigma} = r^{\sigma} + \beta P^{\sigma} V^{\sigma}, \qquad (I - \beta P^{\sigma}) V^{\sigma} = r^{\sigma}.$$

Policy evaluation computes the value of the specified policy by solving this linear system. We will do this for both economic models in the notebook.

From a Given Policy to the Optimal Value

For the finite-state, finite-action models below, let Σ denote the feasible stationary deterministic policies. The **optimal value function** is

$$V^*(s) = \max_{\sigma \in \Sigma} V^\sigma(s).$$

Choose the current action, then follow an optimal policy from the next state. This gives the **Bellman optimality equation**:

$$V^*(s) = \max_{a \in \mathcal{A}(s)} \left\{ r(s,a) + \beta \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^*(s') \right\}.$$

- The first term is the current reward; the second is expected continuation value.
- The action is chosen before the next state is observed. Later actions can respond to the state that is realized.

The McCall Search Model: Setup

An unemployed worker receives a wage offer at the beginning of each period and decides whether to accept it.

- Offers are independent across periods: $w \in \mathcal{W}$, a finite wage set, with probability $q(w)$.
- **Accept:** earn w immediately and keep the job at that wage forever. There is no job separation.
- **Reject:** receive unemployment benefit b now and draw a new offer next period.
- The worker maximizes expected discounted income. In our timing, R_{t+1} is the income generated by the decision at date t.

Accepting gives income today; waiting preserves the opportunity to receive a better offer.

The McCall Search Model as an MDP

The state records employment status and the current offered or accepted wage:

$$\mathcal{S} = \{(\mathrm{U}, w), (\mathrm{E}, w) : w \in \mathcal{W}\}.$$

- At (U, w), choose **accept** or **reject**. At (E, w), the only action is **work**.
- Current rewards are $r((\mathrm{U}, w), \text{accept}) = w$, $r((\mathrm{U}, w), \text{reject}) = b$, and $r((\mathrm{E}, w), \text{work}) = w$.
- The **transition kernel** is

$$\begin{aligned} P((\mathrm{U}, w') \mid (\mathrm{U}, w), \text{reject}) &= q(w'), \\ P((\mathrm{E}, w) \mid (\mathrm{U}, w), \text{accept}) &= 1, \\ P((\mathrm{E}, w) \mid (\mathrm{E}, w), \text{work}) &= 1. \end{aligned}$$

  All other transition probabilities are zero.

A policy σ specifies whether to accept each offer while unemployed.

The McCall Search Model: Recursive Values

Write $V_{\mathrm{U}}^{\sigma}(w) = V^{\sigma}((\mathrm{U}, w))$. Once employed, there is no further choice:

$$V_{\mathrm{E}}(w) = w + \beta V_{\mathrm{E}}(w) = \frac{w}{1-\beta}.$$

For a specified acceptance policy, the **policy Bellman equation** is

$$V_{\mathrm{U}}^{\sigma}(w) = \begin{cases} w + \beta V_{\mathrm{E}}(w), & \sigma(\mathrm{U}, w) = \text{accept}, \\ b + \beta \sum_{w' \in \mathcal{W}} q(w') V_{\mathrm{U}}^{\sigma}(w'), & \sigma(\mathrm{U}, w) = \text{reject}. \end{cases}$$

Allowing the worker to choose gives

$$V_{\mathrm{U}}^{*}(w) = \max \left\{ \underbrace{\frac{w}{1-\beta}}_{\text{accept}}, \quad \underbrace{b + \beta \sum_{w' \in \mathcal{W}} q(w') V_{\mathrm{U}}^{*}(w')}_{\text{reject and continue}} \right\}.$$

The notebook evaluates specified wage-threshold policies and their rollouts.

Consumption and Saving: Setup

A household starts each period with a stock of a consumption good. It chooses how much to consume now and how much to save.

- Saving carries goods into the next period one for one. There is no borrowing, and storage capacity is M.
- After the saving decision, the household receives new income Y_{t+1}, independent across periods and uniform on $\{0, \ldots, B\}$.
- Consumption gives utility $u(c) = c^{\alpha}$, where $0 < \alpha < 1$. The household maximizes expected discounted utility.
- This extends the cake-eating problem: saved goods are replenished by new income, so consumption can continue over time.

Saving reduces current consumption but provides resources for future consumption when income is low.

Consumption and Saving as an MDP

Use whole units of goods. The state is current stock s; the action a is the amount **saved**:

$$\mathcal{S} = \{0, \dots, M + B\}, \qquad \mathcal{A}(s) = \{0, \dots, \min(s, M)\}.$$

Consumption, reward, and the next state are

$$c = s - a, \qquad r(s, a) = (s - a)^{\alpha}, \qquad s' = a + Y'.$$

The **transition kernel** is

$$P(s' \mid s, a) = \begin{cases} 1/(B+1), & a \le s' \le a + B, \\ 0, & \text{otherwise.} \end{cases}$$

- $\sigma(s)$ gives saving at stock s; consumption is $s - \sigma(s)$.
- The state space is closed: saving is at most M, and next income is at most B.

Consumption and Saving: Recursive Values

For a specified saving policy σ,

$$V^{\sigma}(s) = \big(s - \sigma(s)\big)^{\alpha} + \frac{\beta}{B+1} \sum_{y=0}^{B} V^{\sigma}\big(\sigma(s) + y\big).$$

Choosing saving optimally instead gives

$$V^{*}(s) = \max_{a \in \{0,\ldots,\min(s,M)\}} \left\{ (s-a)^{\alpha} + \frac{\beta}{B+1} \sum_{y=0}^{B} V^{*}(a+y) \right\}.$$

- Saving more lowers current utility and changes the distribution of next-period resources.
- In the notebook, a chosen rule σ determines P^{σ} and r^{σ}. Solving $(I - \beta P^{\sigma})V^{\sigma} = r^{\sigma}$ gives its value at every stock.

References and Further Reading

1. **Richard S. Sutton and Andrew G. Barto (2018), Reinforcement Learning: An Introduction. 2nd ed. MIT Press. [Book]**
   - Chapter 3: Finite Markov Decision Processes (Sections 3.3, 3.5 and 3.6)
   - Section 4.1: Policy Evaluation (Prediction)
   - Section 8.10: Rollout Algorithms
2. **Thomas J. Sargent and John Stachurski, QuantEcon. [Online lectures]**
   - Job Search I: The McCall Search Model — The McCall Model
   - Discrete State Dynamic Programming — Discrete DPs
   - Discrete State Dynamic Programming — Example: A Growth Model